CHAPTER 14

Random variables, concentration and the law of large numbers

S62, S64, S73, S74, S77

Before you begin · Distributions and spread

Without a calculator, in six minutes. Answer briefly. Use random variables and inequalities. Then choose the targeted follow-up indicated under the questions.

1. Give the expectation of a distribution taking the values 0 or 1 with probabilities 0.7 and 0.3.

2. Calculate $V(X)$ for a Bernoulli distribution with parameter p.

3. Express $|x-2| \geq 1$ without absolute value.

4. What is the opposite inequality?

5. Calculate the mean of the numbers 1,2,3,4.

6. Is a mean always an integer?

Answers

0,3 ; $p(1-p)$; x at most 1 or at least 3; $1 < x < 3$; 5/2; no.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Weight each of the two values by its probability. **2.** Calculate $E(X^2)-E(X)^2$ for a 0/1 variable; R51. **3.** Draw the numbers at distance at least 1 from 2. **4.** Take the complement of the two half-lines, endpoints excluded. **5.** Add up, then divide by the number of values. **6.** Test the two values 0 and 1. The questions R serve as review material with the teacher, even if their whole-class routine is scheduled later.

Objectives

Calculate the expectation and variance of a sum and of a mean; apply Bienaymé-Chebyshev ; determine a sample size.

Course notes

Sums and transformations

For a finite random variable X, taking the values x_i with probabilities p_i,

$$E(X) = \sum_i x_i p_i, \qquad V(X) = \sum_i (x_i - E(X))^2 p_i = E(X^2) - E(X)^2.$$

The standard deviation is $\sigma(X) = \sqrt{V(X)}$. Expanding the square in $V(aX+b)$, the shift b cancels out and the factor a is squared. For finite random variables X, Y and reals a, b,

$$E(X+Y) = E(X) + E(Y), \quad E(aX+b) = aE(X) + b, \quad V(aX+b) = a^2 V(X).$$

Linearity of expectation does not require independence. If X, Y are independent, $V(X+Y) = V(X) + V(Y)$ (admitted). Without this assumption, the formula cannot be applied. The standard deviation of $aX+b$ is $|a|\sigma(X)$.

Example of dependence. If $Y = X$, then $V(X+Y) = V(2X) = 4V(X)$, and not 2 $V(X)$ as soon as $V(X) \neq 0$.

Proof for the binomial distribution. Write $X = B_1 + \cdots + B_n$ with the independent indicator variables of successes. Each B_i satisfies $E(B_i) = p$, $E(B_i^2) = p$ and $V(B_i) = p - p^2$. By linearity and additivity under independence, $E(X) = np$, $V(X) = np(1-p)$.

Sample and mean

A sample of size n of a distribution is a family of n independent variables with the same distribution. If they have expectation μ and variance σ^2, let $S_n = \sum_{i=1}^{n} X_i$ & $M_n = \frac{S_n}{n}$.

Sum and mean

$$E(S_n) = n\mu, \quad V(S_n) = n\sigma^2, \qquad E(M_n) = \mu, \quad V(M_n) = \frac{\sigma^2}{n}, \quad \sigma(M_n) = \frac{\sigma}{\sqrt{n}}.$$

To divide the standard deviation of the mean by 2, multiply the sample size by 4. The variance of the sum increases; that of the mean decreases.

Concentration

Bienaymé-Chebyshev inequality

For a variable with expectation μ, variance σ^2, and $\delta > 0$,

$$P(|X-\mu| \geq \delta) \leq \frac{\sigma^2}{\delta^2}.$$

A bound greater than 1 remains true but gives no further information; we can replace it with 1.

Proof in the finite setting. In $V(X) = \sum_x (x-\mu)^2 P(X=x)$, keep only the values satisfying $|x-\mu| \geq \delta$. Each square is then at least δ^2. Hence $V(X) \geq \delta^2 P(|X-\mu| \geq \delta)$.

Law of large numbers

Applied to the mean of a sample,

$$P(|M_n - \mu| \geq \delta) \leq \frac{\sigma^2}{n\delta^2} \xrightarrow[n\to\infty]{} 0.$$

This is the law of large numbers in the setting considered: for a fixed precision, the probability of a deviation at least equal to this precision tends to zero. This result does not say that each realisation approaches μ monotonically.

Choosing a sufficient sample size

To ensure $P(|M_n - \mu| < \delta) \geq 1 - \alpha$, it suffices to choose

$$n \geq \frac{\sigma^2}{\alpha\delta^2}.$$

Take the ceiling. It is the smallest integer satisfying this bound, not necessarily the minimum size for the exact probability.

For a success frequency, $\mu = p$ and $\sigma^2 = p(1-p) \leq \frac{1}{4}$. If p is unknown, $n \geq \frac{1}{4\alpha\delta^2}$ is a uniform guarantee.

From the individual distribution to the mean

14.1 Describing a non-binomial variable

First use: S62.

Worked example · A repair cost

A cost X in euros takes the values 0, 20 and 50 with probabilities 0 ,5, 0,3 and 0,2. $E(X) = 16$, $E(X^2) = 620$, therefore $V(X) = 620 - 16^2 = 364$ in euros squared. The std dev is $\sqrt{364} \simeq 19{,}08$ euros. The mean 16 is not one of the possible values of X; it describes a theoretical average over many realisations.

A1 · Immediate application X takes values 1, 2, 4 with probabilities 1 /2, 1/4, 1/4. Check the distribution and calculate the expectation, variance and standard deviation.

14.2 A discount also changes the spread

First use: S62.

Property · Affine transformation

For $Y = aX + b$, $E(Y) = aE(X) + b$ and $V(Y) = a^2V(X)$. The standard deviation is $|a|\sigma(X)$. Adding a constant shifts all values without changing their spread.

Worked example · Cost after a discount

For the previous cost, a 20% discount followed by 5 euros of fees gives $Y = 0{,}8X + 5$. Thus $E(Y) = 17.8$ and $V(Y) = 0{,}64 \times 364 = 232.96$. The variance is neither $0.8V(X) + 5$ nor $0.8\,V(X)$.

A2 · Partial guidance For the distribution in the previous exercise, calculate E and V of $Y = 3X - 2$, then of $Z = -X$. Explain why multiplying by -1 does not change the variance.

14.3 Independence matters for the variance

First use: S64.

Property · Sum and mean

For variables $X_1, \ldots, X_n$ with the same expectation μ and the same variance σ^2, the mean M_n has expectation μ. If they are independent, $V(M_n) = \sigma^2/n$ and $\sigma(M_n) = \sigma/\sqrt{n}$. Linearity of the expectation does not require independence.

Proof. With $S_n = \sum X_i$, linearity gives $E(S_n) = n\mu$. If indep., the admitted additivity of variances gives $V(S_n) = n\sigma^2$. Multiplying by 1 /n for the mean multiplies the expectation by 1 /n and the variance by 1 /n^2. For a Bernoulli variable, E=p and $V = p(1-p)$; the binomial variable, sum of n independent Bernoulli variables, therefore has E=np and $V = np(1-p)$. □

Worked example · Mean of a hundred costs

For a hundred independent costs with the same distribution, expectation 16 and variance 364, the total has expectation 1600 and variance 36400. The average cost has expectation 16 and variance 3.64, hence a standard deviation of about 1.91 euros.

A3 · Counterexample X takes the value 0 or 1 with probability 1 /2 each. Compare the variances of $(X_1 + \cdots + X_n)/n$ for independent copies, then for $X_1 = \cdots = X_n = X$.

14.4 A sufficient guarantee, not an exact probability

First use: S73.

Property · Bienaymé-Chebyshev

For $\delta > 0$, $P(|X - \mu| \geq \delta) \leq V(X)/\delta^2$. For a mean of independent variables, $P(|M_n - \mu| \geq \delta) \leq \sigma^2/(n\delta^2)$. Any bound greater than 1 may be replaced by 1.

Worked example · Size guaranteed by the bound

For a success frequency with probability 1 /2, $V(M_n) = 1/(4n)$. We want a gap strictly less than 0 ,1 with probability at least 0 ,95. It suffices to have $25/n \leq 0{,}05$, so $n \geq 500$. This gives the smallest integer satisfying this sufficient inequality; it does not prove that 499 observations actually fail.

A4 · Immediate application A variable has variance 9. What independent sample size guarantees, by this bound, a deviation of the mean strictly less than 0 ,5 with a probability at least 0.99?

A5 · Compare two pieces of information For $X \sim \mathcal{B}(10; 1/2)$ and $M = X/10$, calculate exactly $P(|M - 1/2| \geq 0{,}3)$ and compare with the Chebyshev bound. Work with the exact integer bounds.

14.5 Several samples, several means

First use: S74 and S77.

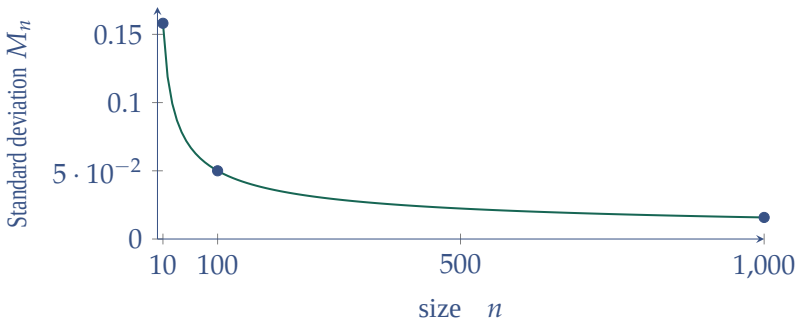


Figure 14.1. Theoretical curve 1/ $(2\sqrt{n})$: multiplying n by 4 divides the standard deviation by 2, not by 4.

Worked example · What is being compared

With n=100, the theoretical standard deviation of a success frequency with probability 1/2 is 0.05. A series of 1,000 experiments of 100 trials produces 1,000 means. It is the standard deviation of this series of means that is compared with 0.05; the 100,000 individual results have a different dispersion.

A6 · Practical work and interpretation With the given program, simulate 1,000 independent samples of sizes 10, 100 and 1,000 for a Bernoulli variable with parameter 1 /2. For each size, compare the observed standard deviation of the means with 1 /(2 $\sqrt{n}$) and count deviations of at least 0 ,1. Why does a single trajectory of frequencies not answer this question?

A7 · Refute "The law of large numbers states that the frequency gets closer to the expectation at each new draw." Give a short sequence of results that temporarily moves the frequency away from 1/2, then restate the result correctly.

One minute, spoken aloud. Write a sentence containing "precision", "risk" and "sufficient size", then distinguish this statement from a certainty for a particular sample.

Delayed review · S90. In a synthesis problem, first identify the distribution or the variance of the mean, before choosing the threshold.

Investigation

Fluctuations of a frequency - 40 min.

A fair coin is tossed independently. For every integer $n \geq 1$, let F_n the frequency of "heads" among the n first tosses.

1. Write a program that simulates 1000 tosses, then plot the points $(n; F_n)$ for $1 \leq n \leq 1000$. Is gap to $\frac{1}{2}$ necessarily decreasing at each toss?
2. Express $E(F_n)$ and $V(F_n)$ as a function of n.
3. Using the Bienaymé-Chebyshev inequality, bound $P(|F_{1000} - \frac{1}{2}| \geq 0{,}05)$.
4. Determine an integer n sufficient to ensure $P(|F_n - \frac{1}{2}| < 0{,}05) \geq 0{,}99$.
5. State what the simulated trajectory shows and what the inequality obtained guarantees. Does a single simulation constitute a proof of this guarantee?

Progressive exercises

E1 · Direct application. Let X a random variable such that $E(X) = 3$ and $V(X) = 2$. Set $Y = 4X - 1$. Calculate the expectation, variance and standard deviation of Y.

E2 · Application and justification. The random variables X and Y are independent and satisfy $E(X) = 2$, $E(Y) = 5$, $V(X) = 1$ and $V(Y) = 4$. Calculate $E(2X - Y)$ and $V(2X - Y)$.

E3 · Method to organise. Consider 64 independent random variables with the same distribution, with expectation 10 and standard deviation 4. Calculate the expectation and standard deviation of their mean.

E4 · Choice of strategy. Let X a random variable such that $E(X) = 100$ and $V(X) = 9$. Using the Bienaymé-Chebyshev inequality, give an upper bound for $P(|X - 100| \geq 6)$.

E5 · Synthesis or proof. Let M_n the mean of n independent random variables with the same distrib., with mean μ and variance 4. Determine a size n sufficient to ensure $P(|M_n - \mu| < 0{,}2) \geq 0{,}95$.

Work from the textbook or a past paper

Baccalauréat, Metropolitan France, 17 June 2025: exercise 1, question 6 (p. 3). At home after S77. Read the context of question 5 and use the values of the expectation and variance given in question 6.
The paper is available here: official past paper . Use the data and instructions of this past paper.

Homework

DM14 - Quality of an estimate - 55 min.

We repeat $n \geq 1$ independent Bernoulli trials with the same parameter $p \in [0;1]$, unknown. We call F_n the frequency of successes.

1. Prove that $E(F_n) = p$ and $V(F_n) = \frac{p(1-p)}{n}$.
2. Show that $p(1-p) \leq \frac{1}{4}$ for all $p \in [0;1]$.
3. Deduce a size n sufficient to ensure $P(|F_n - p| < 0{,}02) \geq 0{,}95$, whatever the value of p.
4. We now know that $p \leq 0{,}1$. Find a sharper upper bound for $p(1-p)$, then improve the sufficient size obtained in question 3.
5. For this question only, the n results are all equal to the same Bernoulli random variable B with parameter p. Calculate F_n and its variance. Explain why the previous guarantee does not apply here.

MCQ A · transformations, sums and means

After S64; MCQ B after S77.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. $E(X) = 3$ and $Y = 2X - 1$. Then $E(Y)$ is:

A. 5 B. 6

C. 2 D. 8

2. $V(X) = 4$ and $Y = -3X + 5$. Then $V(Y)$ is:

A. -12 B. 17

C. 36 D. 41

3. For n independent variables with variance σ^2, the variance of their mean is:

A. $n\sigma^2$ B. σ^2/n

C. σ^2/n^2 D. σ^2

4. The equality $E(X+Y) = E(X) + E(Y)$:

A. requires independence B. requires $X = Y$

C. is false D. holds without independence for finite vars.

5. X is 0 or 2 with probabilities 1/2 each. Its variance is:

A. 0 B. 2

C. 1 D. 4

MCQ B · scope of a guarantee

For each question, only one option is correct. Justify your choice orally before checking the solutions.

6. For $a > 0$, Bienaymé-Chebyshev gives:

A. $P(|X - E(X)| \geq a) \leq V(X)/a^2$ B. $P(|X - E(X)| \geq a) = V(X)/a^2$

C. $P(|X - E(X)| \geq a) \geq V(X)/a^2$ D. $P(X \geq a) \leq E(X)/a^2$

7. For a Bernoulli success frequency, $p = 1/2$, the bound for $P(|F_n - p| \geq 0{,}1)$ is $25/n$. To make it at most 0.05, it is enough to:

A. $n \geq 25$ B. $n \geq 100$

C. $n \geq 250$ D. $n \geq 500$

8. A Chebyshev bound equal to 2 only allows us to deduce:

A. a probability equal to 2 B. the trivial bound 1

C. an impossibility D. a zero probability

9. The law of large numbers states that, for all $a > 0$ fixed:

A. the frequency becomes exactly equal to p B. each new trial reduces the error

C. $P(|F_n - p| \geq a) \to 0$ D. the observations are constant

10. If $X_1 = \cdots = X_n = X$ and $V(X) > 0$, the variance of their mean is:

A. $V(X)$ B. $V(X)/n$

C. 0 D. $nV(X)$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. A sufficient sample size given by Chebyshev is necessarily the true minimum size.

2. The standard deviation of $aX + b$ is $|a|\sigma(X)$.

3. Independence may be omitted when calculating the variance of a mean.

MCQ solutions

1. **A.** By linearity, $E(Y) = 2E(X) - 1 = 5$. The constant is subtracted once.
2. **C.** The variance is multiplied by $(-3)^2 = 9$, and a translation does not change it. It cannot be negative.
3. **B.** The sum has variance $n\sigma^2$, then division by n multiplies the variance by $1/n^2$.
4. **D.** Linearity of expectation does not require independence. Additivity of variances requires additional conditions.
5. **C.** $E(X) = 1$, $E(X^2) = 2$, so $V(X) = 2 - 1^2 = 1$.
6. **A.** It is an upper bound on a deviation from the expectation, not an equality nor a bound applying directly on X.
7. **D.** Solve $25/n \leq 0.05$ with $n > 0$ gives $n \geq 500$. This is a sufficient size according to this bound.
8. **B.** A probability is always at most 1; the upper bound 2 gives no improvement.
9. **C.** This is convergence in probability, not a decrease of the error at each observa-
10. **A.** The mean is exactly X. Repeating the same variable does not create new independent observations.

True/false. False: the upper bound may be crude. True: take the square root of $a^2V(X)$. False: question 10, where the variance does not decrease.

Solutions to the applications and guidance

Application 1. Sum of probabilities 1. E=2; $E(X^2) = 11/2$, therefore $V = 3/2$ and $\sigma = \sqrt{6}/2$.

Application 2. $E(Y) = 4$, $V(Y) = 27/2$; $E(Z) = -2$, $V(Z) = 3/2$. The squared deviations from the mean do not change under a change of sign.

Application 3. Independence: variance 1 $/(4n)$. Identical copies: the mean equals X, so the variance 1/4 whatever n. Repeating the same measurement does not create an independent sample.

Application 4. The bound is 9/ $(0{,}25n) = 36/n$. Requiring $36/n \leq 0.01$ gives $n \geq 3600$.

Application 5. The event is $X \leq 2$ or $X \geq 8$. By symmetry, its probability is 2 $(1 + 10 + 45)/2^{10} = 7/64 \simeq 0{,}109375$. The bound equals $(1/40)/(9/100) = 5/18 \simeq 0{,}2778$, so it is correct but less precise.

Application 6. Theoretical standard deviations are about 0 ,1581, 0,05 and 0,01581. The simulated outputs fluctuate: we are comparing a set of means, not the successive values of a single mean. Empirical proportions are not proofs. The reproducible results file gives the values obtained with the chosen seed.

Application 7. The results 0, 1 give frequency 1 /2; adding 1 gives 2 /3, further from 1 /2. For every fixed precision, the probability of a deviation at least equal to this precision tends to 0 as n tends to infinity. This does not impose a monotonic approach on each realisation.

Reading the guarantees

Have students circle the strict inequality of the central event and the non-strict one of its complement. Always compare a computable probability with a universal bound. Do not present the simulation as a proof of the law of large numbers.

Teaching notes and solutions

Guidance for the teacher

Written test 5 and the second mock exam come before S73: do not ask for Bienaymé-Chebyshev or the law of large numbers. For the variance, have students circle the independence assumption. For the concentration, distinguish exact probability from upper bound, then a sufficient index from the actual minimal index. A simulation is not a proof, and an observed mean does not necessarily get closer with each new trial.

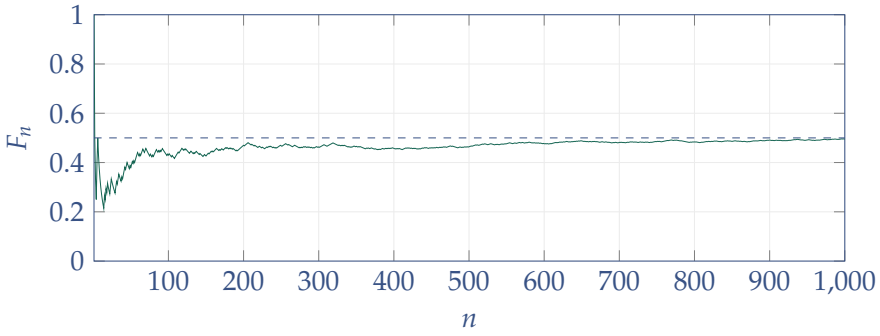
Solutions to the activity

1.

```
from random import Random

def trajectoire_piece(N=1000, graine=2026):
    rng = Random(graine)
    piles, frequences = 0, []
    for n in range (1, N+1):
        piles += rng.random() < 0.5
        frequences.append(piles/n)
    return frequences
```

Plot the points (n, F_n) with the list obtained. After two trials giving one head and one tail, $F_2 = \frac{1}{2}$. Whatever the third outcome, F_3 equals $\frac{1}{3}$ or $\frac{2}{3}$: the gap to $\frac{1}{2}$ increases. The stabilisation is not a monotonic decrease of the deviation along each path.



A simulated path, seed 2026; the line is $y = \frac{1}{2}$.

2. Let B_i the indicator of heads. Then $F_n = \frac{B_1+\cdots+B_n}{n}$. Each B_i has expectation $\frac{1}{2}$ and variance $\frac{1}{4}$. By linearity, $E(F_n) = \frac{1}{2}$; by independence and change of scale, $V(F_n) = \frac{n(\frac{1}{4})}{n^2} = \frac{1}{4n}$.

3. Bienaymé-Chebyshev gives $P(|F_{1000} - \frac{1}{2}| \geq 0{,}05) \leq \frac{1}{4\times 1000\times 0{,}05^2} = 0{,}1$. It is an upper bound, not the exact probability.

4. To ensure a complementary probability of at most 0 ,01, it suffices that $\frac{1}{4n\times 0{,}05^2} \leq 0{,}01$. This is equivalent to $n \geq 10000$. Then $P(|F_n - \frac{1}{2}| < 0{,}05) \geq 0{,}99$.

5. The simulation illustrates one finite realisation; other seeds give other trajectories. The inequality establishes a guarantee on the probability, for any sample satisfying the assumptions of independence and identical distribution.

E1. The affine transformation gives $E(4X-1) = 4E(X) - 1 = 11$ and $V(4X-1) = 4^2V(X) = 32$. The standard deviation is $\sqrt{32} = 4\sqrt{2}$; the shift -1 does not change the spread.

E2. By linearity, $E(2X-Y) = 2E(X) - E(Y) = 4-5 = -1$. The independence of X, Y lets us to add the variances of $2X$ and $-Y$: $V(2X-Y) = 4V(X) + V(Y) = 4+4 = 8$.

E3. The mean of 64 independent identically distributed variables keeps expectation 10. Its variance is $\frac{4^2}{64} = \frac{1}{4}$, so its standard deviation is $\frac{1}{2}$.

E4. Bienaymé-Chebyshev with $\delta = 6$ gives $P(|X-100| \geq 6) \leq \frac{V(X)}{6^2} = \frac{9}{36} = \frac{1}{4}$. This bound does not give an equality or an exact value.

E5. We have $P(|M_n - \mu| \geq 0{,}2) \leq \frac{4}{n \times 0{,}2^2} = \frac{100}{n}$. For this bound to be at most 0,05, it suffices that $n \geq 2000$. The complement then has probability at least 0.95.

Solutions to homework DM14

1. Let B_i the independent success indicators. $F_n = (\frac{1}{n})\sum_{i=1}^{n} B_i$, with $E(B_i) = p$, $V(B_i) = p(1-p)$. Hence

$$E(F_n) = \frac{1}{n}\sum_{i=1}^{n} p = p, \qquad V(F_n) = \frac{1}{n^2}\sum_{i=1}^{n} p(1-p) = \frac{p(1-p)}{n}.$$

2. $p(1-p) = p - p^2 = \frac{1}{4} - (p-\frac{1}{2})^2 \leq \frac{1}{4}$. Equality is attained for $p = \frac{1}{2}$.

3. For every $p \in [0,1]$, $P(|F_n - p| \geq 0{,}02) \leq \frac{1}{4n \times 0{,}02^2}$. It suffices that this bound is at most 0,05, i.e. $n \geq 12500$. Taking the complement, this size guarantees the required probability, uniformly in p.

4. On $[0,0{,}1]$, the function $p \mapsto p(1-p)$ is increasing because its derivative $1-2p$ is positive. So $p(1-p) \leq 0{,}1 \times 0{,}9 = 0{,}09$. Now it suffices that $n \geq \frac{0{,}09}{0{,}05 \times 0{,}02^2} = 4500$. These sizes are sufficient in the sense of the upper bound, not necessarily minimal for the exact probability.

5. If all trials copy the same variable B, then $F_n = \frac{B+\cdots+B}{n} = B$. Hence $V(F_n) = p(1-p)$, without division by n. Adding variances was allowed only because of independence; this assumption disappears here. For $0 < p < 1$, increasing the number of copies brings no new information.

Solutions to the textbook or past-paper work

M_N is the average number of universal donors per sample of 100 people. Linearity gives $E(M_N) = 7.14$; independence gives $V(M_N) = \frac{6{,}63}{N}$.

$$P(|M_N - 7{,}14| \geq 0{,}14) \leq \frac{6{,}63}{0{,}14^2 N}.$$

To make this bound at most 0.05, it is enough to take

$$N \geq \left\lceil \frac{6{,}63}{0{,}05 \times 0{,}14^2} \right\rceil = 6766.$$

By complement, $P(7 < M_N < 7{,}28) \geq 0{,}95$.